

Malaika Sud

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EXPERIENCE

Sailing Stone AI

Software Engineer

San Francisco Bay Area, CA

Dec 2024 – Present

- Built a full-stack, retrieval-augmented (RAG) AI assistant (Rust, Next.js, React, TypeScript) that answers user questions about their database landscape, schema metadata, and migration plans, grounding responses in retrieved product and customer documentation plus the last six conversation turns.
- Engineered a hybrid retrieval pipeline combining pgvector semantic search with PostgreSQL full-text search, weighting semantic and keyword relevance 70/30 and deduplicating document chunks into relevance-ranked prompts.
- Designed an asynchronous job-orchestration backend in async Rust that persists durable conversation state in PostgreSQL and streams model responses to the UI in real time over WebSockets, scoped per tenant and account.
- Developed a reusable TypeScript/React chat workspace shared by the full chat page and an embedded contextual panel, with optimistic updates, streaming message merging, retry and timeout recovery, and sanitized Markdown rendering.
- Built an LLM agent system (DSPy, OpenAI API, pgvector) for task automation & semantic search over team task data.
- Shipped the team's task-management platform (Next.js, React, TypeScript) with real-time updates and predictive velocity modeling; re-architected its backend from FaunaDB to PostgreSQL and automated a Google Sheets-to-database pipeline that eliminated manual data entry.

UC Santa Cruz – Tech4Good Lab

UI Components Lead, Web Developer

Santa Cruz, CA

Aug 2022 – Jun 2023

- Led two 10-person UI teams to build 30+ reusable Angular components (HTML, CSS, JavaScript), improving platform accessibility for 500+ users.

UC Santa Cruz – Society of Asian Scientists & Engineers

Engineering Vice President

Santa Cruz, CA

May 2022 – Jun 2023

- Co-organized SASE WR Conference '22 (1,000+ attendees) and ran monthly workshops for 35+ students; led a weekly CS tutoring program that improved mentees' academic performance.

Stanford University CS Bridge

Section Leader

Stanford, CA

Jun 2021 – Jul 2021

- Taught daily Python lessons and guided 22 students from no coding experience to completed PyGame projects (100% completion rate).

EDUCATION

Georgia Institute of Technology

M.S. in Computer Science | 3.88 GPA

Jan 2024 – Expected Dec 2026

Relevant Coursework: Artificial Intelligence, ML for Trading, Database Systems, Software Architecture & Design

University of California, Santa Cruz

B.S. in Computer Science, with Honors | 3.59 GPA

Sep 2021 – Jun 2023

Relevant Coursework: Computer Systems Design, Computational Models, Database Systems, Web Applications

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Rust, SQL, Java

Frontend: React, Next.js, Angular, Tailwind CSS, Radix UI, HTML/CSS

Backend & Data: Rust (Actix Web, Tokio, SQLx), Node.js, Django, Spring Boot, PostgreSQL, pgvector

AI / ML: OpenAI API, DSPy, RAG, embeddings, semantic & hybrid search, prompt engineering

Tools & Infrastructure: Docker, Git/GitHub, WebSockets, Auth0, Figma

TECHNICAL PROJECTS

HourBank

- Developing a backend-heavy neighborhood time-banking marketplace where users exchange local services through hour-based credits, with a TypeScript monorepo, Dockerized PostgreSQL/PostGIS setup, migration-backed domain schema, shared TypeScript contracts, seed data, and initial NestJS APIs for profiles and listings.

Portfolio Website | <http://www.malaika-sud.com/>

- Built and deployed a responsive React portfolio website showcasing my projects and technical experience across desktop and mobile.